Cybersecurity Challenges in Modern Business Information Systems

[BIS426E] New trends in Business Information Systems

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[2025 – 2026]

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Introduction

Background Information

Cybersecurity in contemporary BIS refers to the multifaceted approach that combines strategic and technical elements to protect computer resources, network structures, and corporate information from any threats of being accessed illegally, disturbed, or destroyed. Originally, the three key aspects that were focused on in cybersecurity were known as the CIA Triad, which implied confidentiality, integrity, and availability; however, today the range of cybersecurity aspects includes digital sovereignty and algorithmic resilience. The "castle-and-moat" paradigm used to be dominant until the advent of cloud computing, telecommuting, and the Internet of Things made it outdated. Today, new cybersecurity approaches rely on data mesh solutions wherein the data ownership is decentralized across business divisions, and ZTA becomes imperative.

Purpose of Research

The first objective of the research is to conduct a thorough assessment of the systemic cybersecurity problems that BIS is experiencing during 2025-2026. The purpose of the research includes:

- Analyzing the history of development of cybersecurity as technology progressed from traditional mainframes to modern "cognitive" technology.
- Applying modern theories such as Cyber Routine Activity Theory and Protection Motivation Theory to assess drivers behind risks.
- Identifying crucial technical characteristics and types of cyber risks.
- Conducting an in-depth evaluation of specific case studies to identify failures in the governance of identity management and cybersecurity of the supply chain.

Significance in the Technological Field

The significance of cybersecurity has transitioned from a localized IT concern to a critical pillar of global economic stability and corporate governance.

1. **Macroeconomic Impact:** Cyber failures can now have an impact on a nation's GDP. In 2025, there was a catastrophic failure by a leading manufacturer, which led to losses of production above $2.5 billion. According to the Bank of England, this had an effect on the growth of the UK economy.
2. **Extreme Financial Risk:** In the US, the mean price for a breach has increased to $10.22 million in 2025 as a result of an additional 9% penalty levied by regulations, as well as rising sophistication in digital forensics within hybrid cloud architectures.

3. **Defensive AI Necessity:** With exfiltration processes four times as fast, offensive AI has made traditional signature defense ineffective. Proficiency in defensive automation technology has become indispensable in today's marketplace.
4. **Executive Personal Liability:** Executive managers can now be personally liable for the effectiveness of cybersecurity risk evaluations under 2026 legislation, making cybersecurity management obligatory among corporate leadership.

Research

History of Cybersecurity: A Detailed Timeline

The history of cybersecurity reflects a dialectical struggle between technological innovation and adversarial exploitation, evolving through several distinct epochs over eight decades:

1940s – The Theoretical Foundations

The logical foundations of cybersecurity were established in 1943 with the development of the first digital computers, such as ENIAC. During this decade, computers were massive, standalone electronic machines, and few people had access to or knowledge of how to use them. Consequently, the risk of external attack was virtually non-existent. However, mathematician John von Neumann published his "Theory of Self-Reproducing Automata," providing the first logical proof that a computer program could replicate itself. This theoretical milestone effectively predicted the emergence of the computer virus decades before the first actual infection occurred.

2. 1950s – Precursors and Phone Phreaking

In the 1950s, while computers remained in their infancy, communication networks began to evolve. The decade was marked by the rise of "phone phreaking," which is considered the precursor to computer hacking. Hackers discovered ways to replicate the frequency tones used by telephone companies to manipulate signals and bypass long-distance billing systems. This era demonstrated the first practical attempts to "trick" a network protocol for unauthorized access, a practice dabbled in by iconic tech figures such as Steve Wozniak and Steve Jobs.

3. 1960s – The Birth of Hacking and ARPANET

The 1960s saw the emergence of the term "hacking," coined by members of the MIT Tech Model Railroad Club to describe the practice of controlling and optimizing complex systems. This decade also laid the groundwork for the modern internet with the development of ARPANET, a project by the U.S. agency ARPA designed for departmental data sharing. In 1967, IBM invited students to test the vulnerabilities of a new computer system, a significant event that is now recognized as a precursor to the professional field of cybersecurity and ethical hacking.

4. 1970s – The First Worm and Antivirus

Cybersecurity monitoring officially began in 1971 when Bob Thomas created "Creeper," the first computer worm. Designed to test the security of ARPANET, Creeper moved between terminals and displayed a message: "I'M THE CREEPER: CATCH ME IF YOU CAN". In response, Ray Tomlinson created "Reaper," a program designed to move across the network to hunt and delete Creeper. Reaper thus became the world's first antivirus program and first self-replicating security software. The decade concluded with the first major arrest for a network intrusion when an attacker was apprehended in 1979 for invading "The Ark" at Digital Equipment Corporation.

5. 1980s – Proliferation of Viruses and Global Standards

The 1980s marked a turning point as computing moved from mainframes to personal desktops. In 1983, William Gibson coined the term "Cybersecurity" in his novel *Neuromancer*. That same year, the U.S. Department of Defense published the **TCSEC Orange Book**, establishing formal criteria for evaluating trusted computer systems. The first PC virus, "Brain," was released in 1986 via infected floppy disks, followed by the Morris Worm in 1988, which infected 10% of the internet within 24 hours. In 1987, the commercial cybersecurity industry was born with the launch of firms like McAfee, Avast, and Sophos.

6. 1990s – The World Wide Web and Encryption

The 1990s witnessed the mass adoption of the internet, which globalized digital vulnerabilities. New threats such as phishing and Distributed Denial-of-Service (DDoS) attacks emerged as hackers began targeting large firms and government departments. To secure the expanding web, Netscape developed the **Secure Sockets Layer (SSL)** protocol in 1995, enabling encrypted online transactions and the foundation of HTTPS. High-profile arrests, such as those of Kevin Mitnick, brought cybercrime into the public consciousness, even inspiring Hollywood films like *Sneakers*.

7. 2000s – Professionalized Cybercrime and State Conflict

The 2000s saw the emergence of highly organized hacking groups, such as Anonymous, and the rise of devastating malware like the ILOVEYOU virus (2000), which caused $15 billion in damages. The decade was characterized by the rise of botnets—infected networks used to perform repetitive tasks or launch massive DDoS attacks. A critical milestone occurred in 2007 when Estonia suffered the first major state-sponsored cyberattack, demonstrating that cyber warfare could disrupt an entire nation's infrastructure and financial services.

7. 2020 – The Remote Work Revolution

The COVID-19 pandemic forced a global shift to telecommuting, expanding the attack surface beyond traditional office perimeters. Cyberattacks on remote workers increased in frequency, and the "Zoom Security Saga" highlighted the difficulty of balancing usability with security under pressure. The year concluded with the SolarWinds attack, a sophisticated supply chain breach where hackers added malicious code to software updates, forcing a massive reevaluation of software trust models.

8. 2021 – Infrastructure and Infrastructure Risk

In 2021, the ransomware group Darkside attacked the Colonial Pipeline, causing gas shortages and price spikes across the U.S. The year also saw the Kaseya supply chain attack and the exploitation of Microsoft Exchange zero-day vulnerabilities (Hafnium) at lightning speed. Academically, the Electronic Arts (EA) breach became a landmark case, demonstrating that a $10 stolen Slack cookie could bypass MFA and lead to the theft of 780GB of source code.

9. 2022 – The Rise of Tech-Giant Targeting

The hacking group **Lapsus$** emerged in 2022, successfully targeting Microsoft, Samsung, and Uber to exfiltrate sensitive data. Nation-state conflicts spilled into the digital domain, with DDoS attacks hitting the Moscow Stock Exchange and Ukrainian banks. Australia experienced its largest data breach with the **Optus** incident, affecting 9.7 million customers, while Costa Rica declared a national state of emergency following a ransomware attack on its Ministry of Finance.

10. 2023 – The API and Ransomware Milestone

Ransomware payments by companies reached a staggering **$1 billion** in 2023. The exploitation of a single vulnerable partner, API, allowed an attacker to scrape **49 million customer records** from Dell, emphasizing that API security had become a primary business risk. The year was also defined by the **MOVE, it** transfers vulnerability, which impacted thousands of organizations worldwide through a single third-party flaw.

11. 2024 – Systemic Dependency and Critical Failures

Early 2024 saw the **CrowdStrike** software update outage, where a faulty update caused widespread system crashes across global client networks, demonstrating the fragility of systemic dependencies. The **Change Healthcare** ransomware attack resulted in a $22 million payment and a total financial impact of $2.87 billion, marking it as one of the most consequential attacks in history. Additionally, the **Salt Typhoon** group compromised major telecom providers to steal call information and government surveillance data.

12. 2025 – The Macroeconomic Siege

In 2025, **Jaguar Land Rover** suffered a catastrophic shutdown of its global production facilities following an attack by "Scattered Spider," costing **$2.5 billion** and impacting UK GDP growth. The **Marks & Spencer** ransomware attack suspended online ordering for six weeks, incurring £300 million in lost profit. The year also witnessed the largest credential-stuffing dump in history, the **16 billion Password Leak**, which fueled a surge in automated account takeovers.

13. 2026 – The Cognitive Frontier (Present)

As of early 2026, the focus has shifted to **Agentic AI**—autonomous systems capable of conducting their own reconnaissance and adjusting payloads without human intervention. The **EU AI Act** becomes fully enforceable by August 2026, carrying fines up to 7% of global revenue for non-compliance. Industrial-scale deepfakes have become a daily reality, illustrated by a **$25.6 million** loss in Hong Kong, where an employee was tricked by an AI-generated video call.

Figure 1: This timeline illustrates the evolution of cybersecurity from the 1960s to the future - starting with password protection and early viruses, advancing through the internet era malware, then shifting to AI-driven defense, quantum threats, and zero-trust models in recent years

Theoretical Framework: 5 Core Academic Theories and Concepts

1. **Protection Motivation Theory (PMT)**

   The Protection Motivation Theory, initially suggested by Rogers (1975), provides reasons for an employee to either accept or reject the security practices. The PMT involves two types of cognitive processes, namely Threat Appraisal and Coping Appraisal.

   Concept - Fear Appeal: The use of threatening messages to trigger the protection motivation process.

2. **General Deterrence Theory (GDT)**

   According to GDT, people are deterred from committing digital offenses based on the fear of sanctions. These include Certainty (likelihood of detection), Severity (penalty is more than the gain), and Celerity (speed of punishment).

   Concept - Specific Deterrence versus General Deterrence: Specific deterrence deals with repeat offenders, while general deterrence serves as an example for others.

3. **Theory of Planned Behavior (TPB)**

   According to TPB, employee intentions to follow organizational policies on cybersecurity can be predicted using attitude towards such action, subjective norms, and perceived behavioral control.

   Concept - Sufficiency Assumption: The notion that the aforementioned three elements suffice in explaining intentions to perform any behavior.

4. **Cyber-Routine Activity Theory (CRAT)**

   Based on the concept of RAT (1979), CRAT provides an analysis of digital convergence crimes based on three factors: a Motivated Offender, a Suitable Target, and No Capable Guardian.

   Definition - Asynchronous Convergence: This is because, unlike physical crimes, where the offenders and the victims have to interact simultaneously in cyberspace.

5. **Situational Crime Prevention (SCP)**

   Target hardening is at the core of SCP, which seeks to manipulate the environment in a manner that eliminates criminal opportunities. SCP applies 25 methods grouped into five approaches: increased effort, increased risk, reduced reward, reduced provocation, and removal of excuses.

   Joint Optimization Concept: Concurrent design of both technical and social systems without compromising efficiency.

Features of Modern Cybersecurity (2025–2026)

Modern cybersecurity systems are characterized by autonomous, identity-driven, and context-aware capabilities:

1. **Cybersecurity Mesh Architecture (CSMA):** CSMA is a composable and scalable approach proposed by Gartner that extends security controls to widely distributed assets. It allows the definition of security perimeters around individual devices or users rather than the entire network, reducing the financial impact of incidents by an estimated **90%**.
2. **Explainable AI (XAI) and Transparent Decision-Making:** As AI takes control of SOC workflows, XAI techniques (e.g., **SHAP, LIME**) provide transparency into "black-box" models. This allows human analysts to understand why an alert was triggered, ensuring trust and regulatory compliance in automated threat response.
3. **Behavioral Biometrics and Continuous Authentication:** Modern systems utilize **Long Short-Term Memory (LSTM) RNNs** to analyze multimodal data, including keystroke dynamics, touch pressure, and device tilt. This provides a passive authentication signal that operates invisibly, detecting anomalies with up to **98.25% accuracy**.
4. **Zero Trust Architecture (ZTA):** A holistic paradigm assuming no implicit trust. It requires continuous, real-time verification of identity and device posture for every access request, regardless of whether it originates from inside or outside the corporate network.
5. **Data Security Posture Management (DSPM+):** DSPM+ tools move beyond mere visibility to active enforcement, automatically discovering sensitive data in cloud environments, applying encryption, and adjusting access controls based on the **intent** of the user.

6. **Micro-segmentation:** This involves dividing the network into granular, isolated zones. If an attacker breaches one segment, micro-segmentation prevents lateral movement, effectively "quarantining" the threat.
7. **Quantum-Resistant Cryptography:** With the threat of quantum computing breaking current encryption by 2030, organizations are transitioning to **NIST-approved quantum-resistant algorithms** to secure long-term data sensitivity.
8. **Automated Incident Response (SOAR):** Systems that use AI-driven orchestration to detect and contain threats in milliseconds. It allows for "site quarantine" or the deactivation of compromised sessions without human intervention.

Types of Cybersecurity Threats

Modern BIS must defend against a diverse spectrum of 13 sophisticated attack types:

1. **Malware:** Malicious software, including viruses, worms, and spyware, designed to infiltrate, damage, or gain unauthorized access to computer systems.
2. **Phishing:** Deceptive communication (email, SMS) used to manipulate users into revealing sensitive data like credentials or financial details.
3. **Insider Threats:** Risks originating from authorized users (employees, partners) who intentionally or accidentally misuse their legitimate access.
4. **Denial of Service (DOS):** Overwhelming a network or system with massive traffic to disrupt service availability for legitimate users.
5. **SQL Injection:** Exploiting web vulnerabilities by injecting malicious code into input fields to manipulate backend databases.
6. **Man-in-the-Middle (MitM):** Attackers intercept and potentially alter communications between two parties to steal data or session tokens.
7. **Zero-Day Exploit:** An attacking software vulnerability that is unknown to the vendor and has no available patch or defensive signature.
8. **Ransomware:** A type of malware that encrypts an organization's files, demanding payment (ransom) in exchange for the decryption key.
9. **Cross-Site Scripting (XSS):** Injecting malicious scripts into web pages viewed by other users to steal session cookies or perform unauthorized actions.
10. **Advanced Persistent Threat (APT):** Prolonged, targeted infiltration campaigns, often nation-state backed, aimed at stealing intellectual property.
11. **Business Email Compromise (BEC):** High-level impersonation of executives or vendors via email to trick employees into making fraudulent transfers.
12. **Cryptojacking:** Unauthorized use of a victim's hardware resources to mine cryptocurrency, leading to performance degradation and high costs.
13. **Botnet Attacks:** Using a network of compromised "zombie" devices to launch massive, automated campaigns like DDoS or credential stuffing.

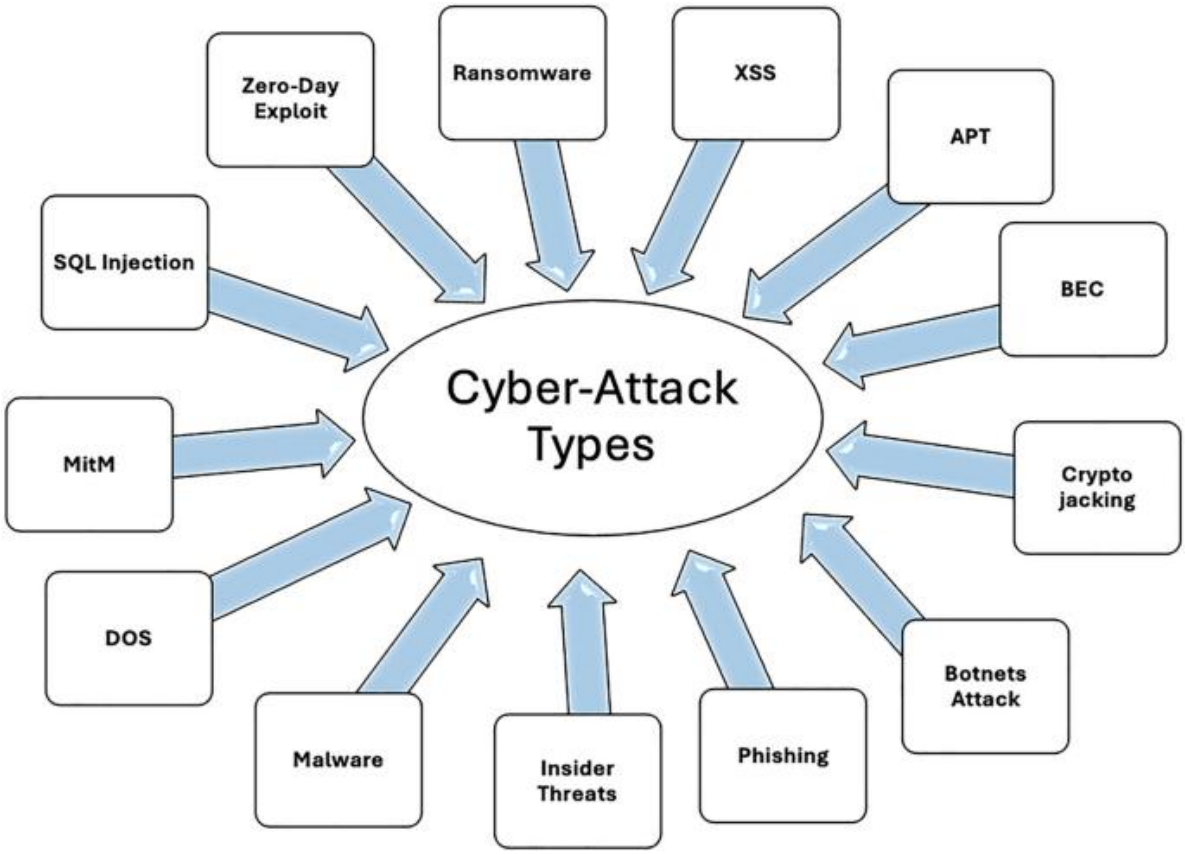

Figure 2: It categorizes common types of cyberattacks, including Zero-Day Exploits, Ransomware, XSS, APT, SQL Injection, BEC, MitM, Cryptojacking, DoS, Botnet Attacks, Malware, Insider Threats, and Phishing - highlighting the diverse methods adversaries use to compromise digital systems.

Current Trends (2025–2026)

- **Converged Operating Models:** Resilient organizations are converging cyber, physical, and operational security into a single unified management model.
- **Regulatory Aggression:** Regulators are shifting from "checkbox" compliance to technical audits and executive personal liability under the EU AI Act and CCPA.
- **Predictive Risk Forecasting:** Incident response is moving upstream, utilizing AI to predict which users or data paths are most likely to trigger future incidents.

Case Studies: Real-World Implementation Analysis

1. Case Study 1: Jaguar Land Rover (JLR) Systemic Supply Chain Resilience (2025)

In late August 2025, the automotive giant Jaguar Land Rover (JLR) fell victim to a Category 3 systemic cyber event orchestrated by the "Scattered Lapsus$ Hunters" group, an alliance of Scattered Spider, LAPSUS$, and Shiny Hunters. The attackers exploited critical unpatched vulnerabilities in third-party SAP NetWeaver software (CVE-2025-31324 and CVE-2025-42999) to deactivate global production line controls and dealer ordering systems. The operational paralysis lasted for five weeks, resulting in a direct financial loss of £196 million and an estimated total UK economic impact of £1.9 billion ($2.5 billion). This incident transformed UK macroeconomic policy, as the Bank of England identified the breach as a contributing factor to slower growth in national GDP. JLR subsequently overhauled its "Plan B" protocols, emphasizing the necessity of manufacturing continuity during total IT-environment shutdowns.

2. Case Study 2: Marks & Spencer (M&S) The Failure of the Human Layer (2025)

The April 2025 ransomware attack on British retailer Marks & Spencer (M&S) serves as a definitive study on identity-centric vulnerability and helpdesk deception. Threat actors gained initial access by impersonating a legitimate M&S employee during a phone call to a third-party IT service desk (Tata Consultancy Services), successfully tricking agents into resetting Active Directory credentials. Once inside, the group exfiltrates the NTDS.dit database file to crack password hashes offline before deploying the Dragon Force ransomware payload. The resulting six-week suspension of online ordering and Click & Collect services cost M&S approximately £300 million in lost profit and wiped £750 million from its market value. The failure underscores the limitations of technical perimeters when human-centric "SIM swapping" and impersonation tactics are employed, leading to the implementation of mandatory, AI-driven "human firewall" simulation training at M&S.

3. Case Study 3: Electronic Arts (EA) Industrial Session Hijacking (Historical Context)

The exfiltration of 780GB of source code from Electronic Arts (EA) remains the most cited academic case for "post-authentication" session hijacking. The breach originated not from a technical exploit, but from a $10 purchase on a dark web marketplace for a stolen Slack authentication cookie. By utilizing the active session token, attackers bypassed the primary login and MFA prompts entirely to enter the company's internal workspace. This foothold allowed them to social engineer a VPN token from support personnel to exfiltrate proprietary source code for FIFA 21 and the Frostbite engine. In response, global BIS standards have pivoted toward Post-Authentication Behavioral Monitoring, utilizing Digital Threat Scoring to identify anomalies within active sessions and automatically terminate them.

4. Case Study 4: Volkswagen (VW) Proactive API Runtime Enforcement (2025)

Security researchers in 2025 identified critical vulnerabilities in the **"MyVW"** mobile application APIs, exposing millions of users to takeover risks. The technical analysis revealed a lack of rate-limiting on one-time passcode (OTP) verification endpoints, which allowed automated brute-force scripts to achieve unauthorized access. Furthermore, the APIs inadvertently returned internal system credentials in plaintext and permitted the exposure of personal owner data via Vehicle Identification Number (VIN) lookups. Volkswagen successfully mitigated the threat through rapid patching and the implementation of Level 4 "Enhanced Trust" APIs, which utilize mutual TLS (mTLS) and certificate-bound tokens. The case highlights the "trusted client" fallacy and the necessity for API security to be elevated as a board-level risk priority.

5. Case Study 5: Hong Kong Multinational — The $25.6M Deepfake Fraud (2025)

A catastrophic deepfake fraud targeting a multinational firm in Hong Kong in 2025 demonstrated the industrial-scale risks of AI-mediated social engineering. An employee was deceived into attending a video conference call in which every other participant, including the Chief Financial Officer, was a high-fidelity AI-generated deepfake mimicking real executive personas. Trusting the visual and auditory authenticity of the deepfakes, the victim authorized several fraudulent transfers totaling $25.6 million. This event signifies the collapse of video and audio as reliable identity verification forms in 2026. It has forced a global shift toward "out-of-band" verification and daily "trust codes" for all sensitive financial transactions.

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Advantages and Disadvantages of Modern BIS Security Models

Advantages of Cybersecurity

- **Protects Sensitive Data:** It is the primary defense for keeping personal information, intellectual property, and financial records safe from unauthorized access and data breaches.
- **Prevents Financial Loss:** By blocking ransomware, fraud, and cyberattacks, it saves organizations and individuals from the massive costs associated with stolen funds, legal penalties, and recovering from a breach.
- **Maintains Business Continuity:** Cybersecurity prevents malicious attacks (like DDoS) from taking down websites and internal networks, ensuring that businesses can stay operational without costly downtime.
- **Builds Trust and Reputation:** Customers are much more likely to do business with companies that have a strong track record of protecting their users' data.
- **Ensures Regulatory Compliance:** Many industries are required by law to protect user data (e.g., GDPR, HIPAA). Proper cybersecurity ensures you meet these legal standards and avoid heavy fines.
- **Defends Against Evolving Threats:** Modern cybersecurity tools actively detect and neutralize a wide variety of threats, from phishing emails and spyware to complex malware and viruses.

Disadvantages (and Challenges) of Cybersecurity

- **High Costs:** Implementing top-tier cybersecurity solutions requires a significant financial investment in software, hardware, and ongoing subscriptions.
- **Need for Constant Updates:** The cyber threat landscape evolves daily. Security systems require constant patching, monitoring, and updating to stay effective against newly invented attack methods.
- **Impact on System Performance:** Heavy security protocols, frequent background scans, and strict firewalls can sometimes slow down network speeds and consume significant system resources.
- **User Inconvenience:** Strict security measures (like mandatory frequent password changes, complex multi-factor authentication, and restricted access rights) can frustrate users and slow down daily productivity.
- **Requires Specialized Talent:** There is a global shortage of highly trained cybersecurity professionals. Hiring and retaining experts to manage and monitor these systems is difficult and expensive.

- **Creates a False Sense of Security:** No system is 100% foolproof. Relying too heavily on software can lead to human complacency, making organizations vulnerable to social engineering or phishing attacks that target human psychology rather than software flaws.

Conclusion:

The evolution of cybersecurity within modern Business Information Systems (BIS) has shifted from static "castle-and-moat" defenses to a decentralized, data-centric model. Traditionally, security focused on the "CIA Triad": confidentiality, integrity, and availability, but the 2025–2026 landscape now emphasizes digital sovereignty, algorithmic resilience, and Zero Trust Architecture (ZTA). This transition is driven by the obsolescence of traditional perimeters due to cloud computing and the Internet of Things, moving the industry toward "data mesh" solutions where security is embedded into decentralized business divisions.

The current threat environment is characterized by "industrialized" and AI-mediated attacks that have rendered many legacy protections ineffective. Organizations now face Agentic AI systems capable of autonomous reconnaissance and sophisticated "Session Hijacking" tactics that bypass standard Multi-Factor Authentication (MFA) by harvesting active session cookies. These technical threats are compounded by "Social Engineering 2.0," which utilizes high-fidelity deepfakes to deceive employees, as seen in the 2025 Hong Kong case, where a firm lost $25.6 million to an AI-generated video conference.

To mitigate these risks, modern BIS security models are integrating autonomous, context-aware capabilities like User and Entity Behavior Analytics (UEBA) and Automated Incident Response (SOAR). Beyond technical implementation, there is a significant shift in corporate governance, where cybersecurity has become a critical pillar of macroeconomic stability and executive liability. Under 2026 legislation, executive managers can now be held personally liable for the effectiveness of risk evaluations, reflecting a global trend where a single systemic cyber event, such as the £1.9 billion impact of the 2025 Jaguar Land Rover breach, can affect a nation's total GDP.

References

- Unit 42 Incident Response Report: https://www.paloaltonetworks.com/resources/research/unit-42-incident-response-report
- Top 10 Cybersecurity Risks:

  https://arctiq.com/blog/top-10-cybersecurity-risks-of-2026
- 10 Cyber Security Trends For 2026:

  https://www.sentinelone.com/cybersecurity-101/cybersecurity/cyber-security-trends/
- Modern API Attacks Threat Stats:

  https://securityboulevard.com/2026/02/inside-modern-api-attacks-what-we-learn-from-the-2026-api-threatstats-report/
- AI-Powered Scams 2026:

  https://www.eccu.edu/blog/social-engineering-attacks-generative-ai-2026/
- Social Engineering Defense:

  https://www.nucamp.co/blog/social-engineering-in-2026-the-real-world-scams-and-how-to-defend-against-them
- Remote Work Impact:

  https://360advanced.com/the-impact-of-remote-work-on-security-and-compliance/
- Breach Costs Trend:

  https://www.auxis.com/10-cybersecurity-trends-defining-2026/
- API Security Guide 2026:

  https://www.raidiam.com/api-security-the-definitive-guide-for-2026-and-beyond
- Rise of Session Hijacking: https://brandefense.io/blog/mfa-doesnt-protect-you-cookies-give-you-away-the-rise-of-session-hijacking/
- EU AI Act Compliance:

  https://www.digitalapplied.com/blog/eu-ai-act-2026-compliance-us-businesses-must-do-now
- Top Breaches of 2025:

  https://www.int-comp.org/insight/top-10-cybersecurity-breaches-of-2025-lessons-for-compliance/
- History of Cyber Security:

  https://www.geeksforgeeks.org/computer-networks/history-of-cyber-security/